

AYMANE AIT DADS

AI/ML Engineering Intern | Machine Learning · Data Analysis · Generative AI Prototyping

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PROFESSIONAL SUMMARY

End-to-end ML pipeline built at Orange Maroc on 100K+ fiber-optic network records using Random Forest and K-Means clustering reduced manual analysis time by ~60% and directly informed network optimization decisions. Top 10% public leaderboard ranking (0.80/1.0) in the NVIDIA Nemotron Reasoning Challenge (\$106K+ prize pool) demonstrates applied machine learning prototyping and rigorous ablation-driven model evaluation. Proficient in Python, NumPy, Pandas, Scikit-learn, PyTorch, and Hugging Face Transformers with hands-on experience in machine learning, data analysis, automation, and Generative AI applications. Available May/June 2026 for a full-time, in-office summer internship in Mississauga, Ontario, Canada; holds valid documentation to work in Canada.

CORE COMPETENCIES

Machine Learning | Data Analysis | Generative AI | Python Programming | Prototyping & Testing | Automation | Problem-Solving | Software Development

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built and evaluated end-to-end machine learning pipeline combining Random Forest classification and K-Means clustering on **100K+ fiber-optic network records** (upload speed, latency, jitter) to classify network quality across thousands of nodes.
- Developed automated feature engineering and model evaluation workflows that reduced **manual analysis time by ~60%**, directly accelerating the network optimization team's maintenance cycle.
- Prototyped a customer performance clustering pipeline mapping profiles to **5 commercial service tiers**, with outputs adopted directly by the network optimization team for production deployment.
- Delivered stakeholder presentations translating model outputs into actionable maintenance recommendations, using **Power BI** dashboards to communicate data analysis findings to non-technical audiences.

PROJECTS

NVIDIA Nemotron Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Top 10% Public Leaderboard

- Achieved **top 10% public leaderboard score (0.80/1.0)** in the \$106K+ prize pool competition by fine-tuning a 30B Mamba-Transformer MoE model with LoRA (r=32, ~888M trainable parameters) via SFTTrainer on 7,828 chain-of-thought examples.
- Resolved 5 Blackwell GPU (sm_120) incompatibilities via systematic CUDA and Triton debugging, and conducted **20+ single-variable ablations** (packing, loss masking, data quality) to confirm each configuration change's isolated impact on leaderboard score.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · 1st in Class Leaderboard

- Ranked **1st on EURECOM private class Kaggle leaderboard (0.791 AUC)** by fine-tuning a CLIP ViT-L/14 backbone (428M params) with a custom MLP head, dual learning rates, and a 3-model ensemble with 10-view TTA across 250K training images.
- Submitted prototype to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; identified through ablation that 1-epoch training (0.793 AUC) outperforms 4-epoch training (0.767 AUC) due to improved out-of-distribution generalization.

PyTorch · OpenCLIP · Hugging Face Transformers · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Statistics, Cloud Computing, Web Applications, Image Security

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML Frameworks & Engineering: Python | PyTorch | Scikit-learn | Hugging Face Transformers | TRL | PEFT | Unsloth

Data Analysis & Visualization: Pandas | NumPy | Matplotlib | Seaborn | Power BI | feature engineering | EDA

AI/GenAI & MLOps: LoRA | QLoRA | vLLM | prompt engineering | Git | Linux | FP16 mixed precision | CUDA debugging